

K8S-04: Cluster Health Monitoring

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Deep-Dive into Kubernetes Cluster Metrics

Cluster health monitoring provides visibility into the infrastructure layer of Kubernetes: nodes, control plane, and cluster-wide resources. This notebook covers key metrics, thresholds, and DQL queries for proactive cluster management.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Kubernetes monitoring
DynaKube	ActiveGate with <code>kubernetes-monitoring</code> capability

Requirement	Details
Permissions	<code>metrics.read</code> , <code>entities.read</code> , <code>logs.read</code> , <code>events.read</code>
Data	At least 24 hours of cluster data

1. Cluster Health Overview

Key Health Indicators

Category	Metrics	Healthy State
Node Status	Ready/NotReady	All nodes Ready
Pod Scheduling	Pending pods	No long-pending pods
Resource Pressure	CPU/Memory pressure	No pressure conditions
Disk Pressure	Disk space, inode usage	>15% available
Network	CNI health, DNS latency	<100ms DNS resolution

Dynatrace Kubernetes Dashboard

The built-in Kubernetes dashboard provides:

- Cluster overview with node status
- Namespace resource usage
- Workload health summary
- Recent events and problems

Navigate to: **Infrastructure > Kubernetes**

Release Radar (April 2026): Enhanced Kubernetes Visibility

Three additions to the Kubernetes app's cluster and workload views:

Capability	What it surfaces	Why it matters
Horizontal Pod Autoscaler (HPA)	HPA is now a first-class object — scaling triggers, current/desired replica counts, and the workloads it drives	See <i>why</i> a workload scaled (which metric crossed which threshold) without leaving Dynatrace

Capability	What it surfaces	Why it matters
Custom Resources (CRs)	Monitor up to 5 Custom Resources per cluster , surfacing CRD-heavy ecosystems (Argo, Istio, Cert-Manager, Kyverno, operator-managed databases)	Brings operator/CRD state into the same view as native Kubernetes objects
Cloud configuration in cluster details	The underlying managed-cluster configuration (EKS, AKS, GKE) shown inline as YAML or JSON	Correlate cluster and cloud state in one place — no jumping to the cloud console

Note: The per-cluster Custom Resource cap (5) means you choose which CRDs matter most — prioritize the operators whose state actually drives incidents in your environment.

Kubernetes Enhanced Object Visibility (ActiveGate 1.327+)

Building on the above, the Kubernetes app now surfaces a broader set of objects and their raw definitions:

- **Additional Kubernetes objects** — Ingress, NetworkPolicies, CRDs, PVCs, PVs, ConfigMaps, and more appear alongside the native cluster/node/namespace/workload views.
- **YAML definitions inline** — view an object's YAML to debug and validate configuration in real time without leaving Dynatrace.
- **Query YAML across clusters with DQL** — surface misconfigurations, missing references, or policy violations across all clusters and namespaces at once.

Prerequisite: ActiveGate version 1.327+. Older ActiveGate versions stay in backward-compatibility mode, where an extra **Explorer (Classic)** tab appears. From June 2026, Explorer Classic transitions to *maintenance-only* support — upgrade ActiveGate to 1.327+ to move clusters to the new Explorer before automatic migration. No monitoring data is lost during the transition.

Kubernetes Connection Lifecycle — Automatic Stale Cleanup (SaaS 1.344)

Every monitored cluster has a **Kubernetes connection** on the tenant side. Its lifecycle is worth knowing in both directions, because it cuts two ways.

SaaS 1.344 (released 07/27/2026, **staged tenant rollout from 07/29/2026** — verify it has reached your tenant before relying on it) adds **automatic cleanup of stale Kubernetes connections at a 60-day threshold**:

Direction	What happens	What you should do
Decommissioned cluster	Its connection stops reporting and is now cleaned up automatically once it crosses 60 days stale — the connection list stops accumulating entries for clusters that no longer exist.	Nothing. This is the behavior you want; it removes a recurring housekeeping chore.
Live but long-idle cluster	A cluster that is still real but has not reported for 60 days — a lab, a seasonal environment, a cluster whose ActiveGate has been down a long time — can be reaped by the same rule , and reconnecting means re-establishing the connection.	Treat "no data for weeks" as an issue to fix rather than a state to tolerate. If an environment is legitimately dormant for months, expect to re-add it and note that in your runbook.

Until 1.344 reaches your tenant, stale connections persist until someone removes them manually — the existing housekeeping step remains the working path, and reviewing the connection list periodically is still worth doing.

SaaS 1.344 also adds **cross-app navigation with context preservation**, so a jump from the Kubernetes app into another app (for example, into logs or a dashboard) carries the cluster/namespace context with it instead of dropping you at an unfiltered start.

```
// List all monitored Kubernetes clusters (smartscape topology)
smartscapeNodes "K8S_CLUSTER"
| fields entity.name = name, tags
| sort entity.name asc

// Legacy alternative (deprecated for new content):
// fetch dt.entity.kubernetes_cluster
// | fields entity.name, tags
// | sort entity.name asc
```

2. Node Monitoring

Node Status and Conditions

Condition	Description	Alert When
Ready	Node can accept pods	False for >5 min

Condition	Description	Alert When
MemoryPressure	Low memory	True
DiskPressure	Low disk space	True
PIDPressure	Too many processes	True
NetworkUnavailable	Network not configured	True

```
// List all Kubernetes nodes (smartscape topology)
smartscapeNodes "K8S_NODE"
| fields entity.name = name, tags
| sort entity.name asc

// Legacy alternative (deprecated for new content):
// fetch dt.entity.kubernetes_node
// | fields entity.name, tags
// | sort entity.name asc
```

```
// Node CPU utilization – container usage summed per node, against node
allocatable
// There is no dt.kubernetes.node.cpu_usage metric: node-level usage is derived
// by summing the per-container metric. Allocatable is a genuine node-grain
metric.
timeseries {
    used = sum(dt.kubernetes.container.cpu_usage),
    allocatable = avg(dt.kubernetes.node.cpu_allocatable)
}, from:-1h, by:{k8s.cluster.name, k8s.node.name}
| fieldsAdd cpuPercent = round(100 * arrayAvg(used) / arrayAvg(allocatable),
decimals: 1)
| fields k8s.cluster.name, k8s.node.name, cpuPercent
| sort cpuPercent desc
| limit 10
```

```
// Node memory utilization – working-set memory summed per node, against node
allocatable
// Same derivation as CPU above: no dt.kubernetes.node.memory_usage metric
exists.
timeseries {
    used = sum(dt.kubernetes.container.memory_working_set),
    allocatable = avg(dt.kubernetes.node.memory_allocatable)
}, from:-1h, by:{k8s.cluster.name, k8s.node.name}
| fieldsAdd memPercent = round(100 * arrayAvg(used) / arrayAvg(allocatable),
decimals: 1)
| filter memPercent > 80
| fields k8s.cluster.name, k8s.node.name, memPercent
| sort memPercent desc
```

```
// Node filesystem usage - disk pressure detection
timeseries avgDiskUsage = avg(dt.host.disk.used.percent), from:-1h, by:
{dt.entity.host}
| fieldsAdd avgDiskUsageValue = arrayAvg(avgDiskUsage)
| filter avgDiskUsageValue > 80
| sort avgDiskUsageValue desc
```

3. Resource Capacity Planning

Capacity Metrics

Metric	Description	Use Case	Grail metric key
Allocatable	Resources available for pods	Scheduling decisions	<code>dt.kubernetes.node.cpu_allocatable</code> , <code>dt.kubernetes.node.memory_allocatable</code>
Requested	Sum of pod requests	Capacity planning	<code>dt.kubernetes.container.requests_cpu</code> , <code>dt.kubernetes.container.requests_memory</code>
Used	Actual consumption	Right-sizing	<code>dt.kubernetes.container.cpu_usage</code> , <code>dt.kubernetes.container.memory_working_set</code>
Limits	Maximum allowed	Burst capacity	<code>dt.kubernetes.container.limits_cpu</code> , <code>dt.kubernetes.container.limits_memory</code>

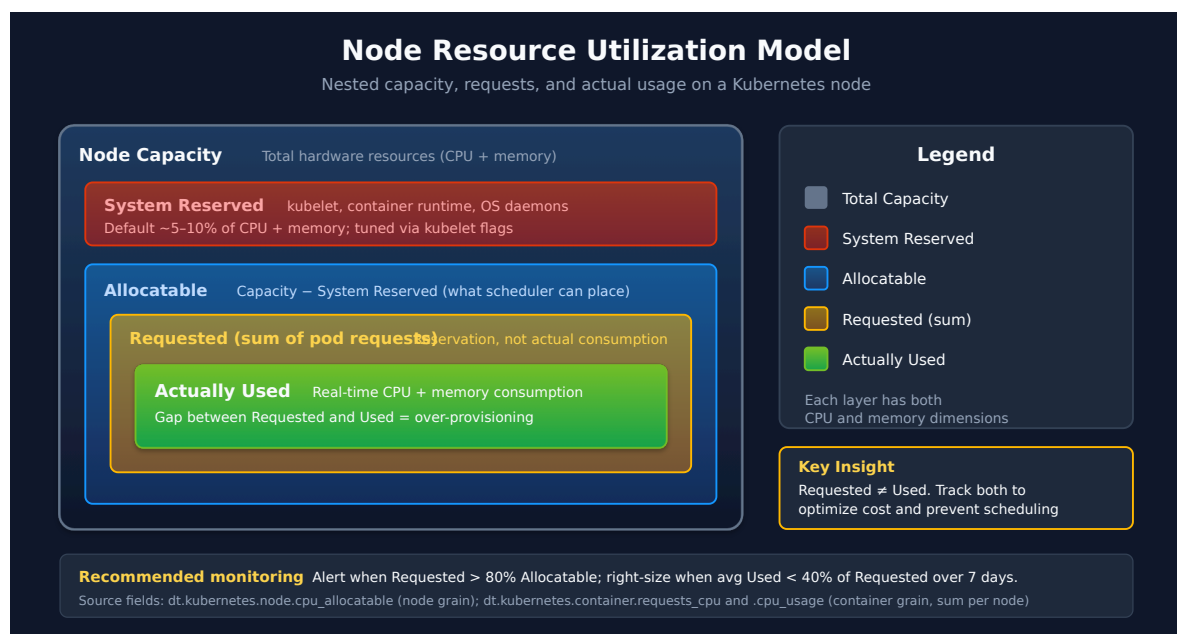
Grain matters more than the label. *Allocatable* is the only genuinely node-grain family — everything else is emitted **per container** and rolled up by you. There is no

```
dt.kubernetes.node.cpu_usage , no dt.kubernetes.node.memory_usage , and no
dt.kubernetes.workload.requests_* ; node and workload views are derived by
summing the dt.kubernetes.container.* series across the dimensions you group by.
Enumerate what your tenant actually carries before writing a query:
```

```
dql &gt; metrics &gt; | filter startsWith(metric.key, "dt.kubernetes") &gt; |
summarize n = count(), by:{metric.key} &gt; | sort metric.key asc &gt;
```

Note the `metrics` command takes `from:` with **no leading comma** (`metrics from:-2h | ...`). Writing `metrics, from:-2h` is a parse error, and a parse error read only for its record count looks exactly like "this tenant has no such metrics."

Utilization vs. Allocation



Capacity planning across an ActiveGate 1.343 upgrade: the request and limit metrics — `dt.kubernetes.container.requests_cpu / requests_memory` and `limits_cpu / limits_memory` — **include init containers from ActiveGate 1.343 onward**, so the *Requested* line steps up at the upgrade while *Used* (`dt.kubernetes.container.cpu_usage , memory_working_set`) does not move. Read a trend spanning that boundary as **two series, not one** — otherwise the discontinuity reads as a genuine reservation increase and skews right-sizing conclusions. Full explanation: K8S-08 §2.

```
// CPU requests by namespace – requests are a container-grain metric, summed to
namespace
// (there is no dt.kubernetes.workload.requests_cpu; requests live under
container.*)
timeseries cpuReq = sum(dt.kubernetes.container.requests_cpu), from:-1h, by:
{k8s.namespace.name}
| fieldsAdd avgReqMillicores = round(arrayAvg(cpuReq), decimals: 0)
| fields k8s.namespace.name, avgReqMillicores
| sort avgReqMillicores desc
| limit 15
```

```
// Memory requests by namespace (GiB) – container-grain metric summed to
namespace
timeseries memReq = sum(dt.kubernetes.container.requests_memory), from:-1h, by:
{k8s.namespace.name}
| fieldsAdd avgReqGiB = round(arrayAvg(memReq) / 1073741824, decimals: 2)
| fields k8s.namespace.name, avgReqGiB
| sort avgReqGiB desc
| limit 15
```

```
// Find over-provisioned workloads (low CPU usage)
timeseries avgCpuUsageMillicores = avg(dt.kubernetes.container.cpu_usage),
from:-1h, by:{dt.entity.cloud_application}
| fieldsAdd avgCpuUsageMillicoresValue = arrayAvg(avgCpuUsageMillicores)
| sort avgCpuUsageMillicoresValue asc
| limit 20
```

4. Control Plane Health

Control Plane Components

Component	Function	Key Metrics
API Server	REST API for K8s	Request latency, error rate
etcd	Distributed KV store	Disk sync latency, leader elections
Scheduler	Pod placement	Scheduling latency, failures
Controller Manager	Reconciliation loops	Queue depth, sync latency

Managed Kubernetes Note

For managed Kubernetes (EKS, AKS, GKE), control plane metrics are limited. Focus on:

- API server response times (client-side)
- Kubernetes events for scheduling issues
- Cloud provider metrics for control plane health

```
// Node-level and control-plane events
// Data object corrected 08/12/2026. Kubernetes events are NOT logs. This cell
scraped
// `fetch logs` for `log.source` containing "kubernetes" or for content
substrings like "BackOff" –
// no log.source matches, and kubelet event text is not in the log stream, so it
returned nothing
// while the cluster emitted 231,296 Kubernetes events in the same window.
// They arrive as `fetch events | filter event.provider == "KUBERNETES_EVENT"`,
STRUCTURED:
//   dt.kubernetes.event.reason           Unhealthy · BackOff · Killing ·
FailedScheduling ·
//                                       FailedMount · BackoffLimitExceeded ·
EvictionThresholdMet ...
//   dt.kubernetes.event.message           the human-readable text
//   dt.kubernetes.event.important         "true" marks the warning-class events
//   dt.kubernetes.event.involved_object.kind / .name
//   dt.kubernetes.event.count / .first_seen / .last_seen
//   plus k8s.cluster.name · k8s.namespace.name · k8s.pod.name ·
k8s.workload.name · k8s.node.name
// NOTE: event.type is CUSTOM_INFO on every one of these – it is NOT "Warning";
severity lives in
// dt.kubernetes.event.important. Enumerate reasons with:
//   fetch events, from:-24h | filter event.provider == "KUBERNETES_EVENT"
//   | summarize n = count(), by:{dt.kubernetes.event.reason} | sort n desc
fetch events, from:-6h
| filter event.provider == "KUBERNETES_EVENT"
| filter isNotNull(k8s.node.name)
| summarize events = count(), by:{k8s.node.name, dt.kubernetes.event.reason}
| sort events desc
| limit 25
```

5. Cluster-Wide Events

Where Kubernetes events actually live

Kubernetes events are ingested into the `events` object, not into `logs`. Two discriminators are easy to get wrong, and both fail silently:

Getting it wrong	What actually happens
<pre>fetch events \ filter event.kind == "K8S_EVENT"</pre>	<code>event.kind</code> never takes that value . Over 24 h on a live tenant (08/11/2026) the only values present were <code>DAVIS_EVENT</code> , <code>SYNTHETIC_EVENT</code> , <code>FLEET_EVENT</code> and <code>DAVIS_PROBLEM</code> . Kubernetes events carry <code>event.kind == "DAVIS_EVENT"</code> , so <i>kind</i> cannot discriminate them. Zero rows, no error.
<pre>fetch logs \ filter matchesPhrase(log.source, "kubernetes")</pre>	Zero rows after scanning 46.5 GB (same tenant, same window). Expensive and empty — the worst combination.
<pre>k8s.event.reason</pre>	The field resolves but is null on every Kubernetes event record . Another silent zero.

The correct discriminator is `event.provider == "KUBERNETES_EVENT"`, and the reason field is `dt.kubernetes.event.reason`. Companion fields on the same records:

```
dt.kubernetes.event.message, .count, .first_seen, .last_seen,
.involvement.kind, .involved_object.name, plus the standard k8s.cluster.name /
k8s.namespace.name / k8s.pod.name / k8s.workload.name / k8s.node.name dimensions.
```

Event Reasons to Monitor

Reasons observed on tenant `yhu28601` over 24 h (08/11/2026) — your distribution will differ, so run the summary query below against your own estate rather than assuming this shape:

Reason	Count (24 h)	Action
Unhealthy	2,084	Readiness/liveness probe failing — check the probe and the container
BackOff	586	Container restart loop — check logs and exit codes

Reason	Count (24 h)	Action
FailedScheduling	433	No node satisfies requests/affinity/taints
Killing	250	Normal termination during rollout, or eviction
KernelReady	136	Node lifecycle, informational
Failed / FailedMount / BackoffLimitExceeded	tens	Image pull, volume, and Job failures

```
// Event summary by reason – run this first to see what your estate actually
emits
fetch events, from:-24h
| filter event.provider == "KUBERNETES_EVENT"
| summarize eventCount = count(), by:{dt.kubernetes.event.reason}
| sort eventCount desc
| limit 25
```

```
// Scheduling and placement failures – the pods that could not be placed
fetch events, from:-24h
| filter event.provider == "KUBERNETES_EVENT"
| filter dt.kubernetes.event.reason == "FailedScheduling"
| fields timestamp, k8s.cluster.name, k8s.namespace.name,
      dt.kubernetes.event.involved_object.kind,
      dt.kubernetes.event.involved_object.name,
      dt.kubernetes.event.message
| sort timestamp desc
| limit 30
```

```
// OOM kills by workload – the metric, not an event
// OOMKilled has no reliable Kubernetes event reason; Dynatrace emits it as a
counter
// derived from the container status, written only when at least one kill
occurred.
timeseries oom = sum(dt.kubernetes.container.oom_kills), from:-24h,
  by:{k8s.cluster.name, k8s.namespace.name, k8s.workload.name}
| fieldsAdd oomTotal = arraySum(oom)
| filter oomTotal > 0
| fields k8s.cluster.name, k8s.namespace.name, k8s.workload.name, oomTotal
| sort oomTotal desc
| limit 30
```

```
// Which namespaces and workloads generate the most event noise
fetch events, from:-24h
| filter event.provider == "KUBERNETES_EVENT"
| summarize eventCount = count(),
  by:{k8s.cluster.name, k8s.namespace.name, k8s.workload.name,
dt.kubernetes.event.reason}
| sort eventCount desc
| limit 20
```

6. Cost Optimization Queries

Resource Efficiency Analysis

Metric	Target	Action If Not Met
CPU Utilization	>40% avg	Reduce requests
Memory Utilization	>50% avg	Reduce requests
Node Utilization	>60%	Scale down nodes
Idle Pods	0	Review necessity

```
// Find workloads with very low CPU utilization (candidates for right-sizing)
timeseries avgCpuUsageMillicores = avg(dt.kubernetes.container.cpu_usage),
from:-1h, by:{dt.entity.cloud_application}
| fieldsAdd avgCpuUsageMillicoresValue = arrayAvg(avgCpuUsageMillicores)
| sort avgCpuUsageMillicoresValue asc
| limit 25
```

```
// Memory usage efficiency by workload (low usage = over-provisioned)
timeseries avgMemUsageBytes = avg(dt.kubernetes.container.memory_working_set),
from:-1h, by:{dt.entity.cloud_application}
| fieldsAdd avgMemUsageBytesValue = arrayAvg(avgMemUsageBytes)
| sort avgMemUsageBytesValue asc
| limit 25
```

7. Dynatrace Component Health

Monitor the health of Dynatrace's own components (OneAgent, ActiveGate) running on your clusters.

Why Monitor the Monitoring?

Component	What to Watch	Action Threshold
OneAgent	CPU usage, memory consumption	Sustained memory growth above baseline
ActiveGate	Memory headroom, pod restarts	Memory headroom < 20%, any OOMKill
CSI Driver	Volume mount failures	Any mount timeout
Operator	Reconciliation errors	Failed CR updates

Note: OneAgent runs without resource limits or requests by default. Headroom queries using `limits_memory` will return no data for OneAgent containers. Use absolute memory usage instead. ActiveGate **does** have limits configured, so headroom queries work for AG.

Tip: These queries use `matchesValue(k8s.container.name, "dynatrace-oneagent")` to isolate Dynatrace components from application workloads.

```
// OneAgent CPU usage across clusters (top 20 consumers)
timeseries by:{k8s.cluster.name, k8s.pod.name}, from:-1h,
  oaCpu = avg(dt.kubernetes.container.cpu_usage),
  filter:{matchesValue(k8s.container.name, "dynatrace-oneagent")}
| fieldsAdd avgCpu = arrayAvg(oaCpu)
| sort avgCpu desc
| limit 20
```

```
// OneAgent memory usage across clusters (absolute usage – OA has no resource
limits by default)
timeseries by:{k8s.cluster.name}, from:-1h,
  memUsage = avg(dt.kubernetes.container.memory_working_set),
  filter:{matchesValue(k8s.container.name, "dynatrace-oneagent")}
| fieldsAdd avgUsageMi = round(arrayAvg(memUsage) / 1048576, decimals: 0)
| sort avgUsageMi desc
```

```
// Dynatrace component events (last 24h) – restarts, back-offs, scheduling
failures
// Discriminator is event.provider, NOT event.kind (these records carry
event.kind == "DAVIS_EVENT").
fetch events, from:-24h
| filter event.provider == "KUBERNETES_EVENT"
| filter k8s.namespace.name == "dynatrace"
| summarize eventCount = count(), by:{k8s.cluster.name, k8s.workload.name,
dt.kubernetes.event.reason}
| sort eventCount desc
| limit 50
```

8. Alerting Strategies

Recommended Alerts

Alert	Condition	Severity
Node NotReady	Node condition != Ready for 5 min	Critical
High Node CPU	CPU > 85% for 15 min	Warning
High Node Memory	Memory > 90% for 10 min	Critical
Disk Pressure	Disk > 85%	Warning
Pod Scheduling Failed	FailedScheduling events	Warning
OOM Kills	OOMKilled events	Warning

Alert Configuration in Dynatrace

Navigate to: **Settings > Anomaly detection > Kubernetes**

Configure:

- Node availability alerts
- Resource saturation thresholds
- Workload health anomalies

Custom Metric Events

For advanced alerting, use custom metric events with DQL-derived thresholds.

Next Steps

With cluster health monitoring in place, proceed to:

Next Notebook	Topic
K8S-05: Workload Monitoring	Application-level observability
K8S-06: Namespace Organization	Boundaries and access control
K8S-07: Events and Logs	Log ingestion and analysis

Summary

In this notebook, you learned:

- Cluster health overview and key indicators
 - Node monitoring for CPU, memory, and disk
 - Resource capacity planning with requests vs. usage analysis
 - Control plane health considerations
 - Cluster-wide event monitoring and analysis
 - Cost optimization queries for right-sizing
 - Alerting strategies for proactive cluster management
-

References

- [Set up Dynatrace on Kubernetes \(DT docs\)](#)
 - [How K8s monitoring works \(DT docs\)](#)
 - [Full observability deployment \(DT docs\)](#)
 - [Kubernetes app — clusters and workloads view \(DT docs\)](#)
 - [Davis Problems app \(DT docs\)](#)
 - [smartscapeNodes command \(DT docs\)](#)
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